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BSH GUIDELINE

The diagnosis and management of primary mediastinal B-cell lymphoma: A British Society for Haematology guideline

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Summary
Primary mediastinal B-cell lymphoma (PMBCL) is a distinct clinicopathological entity within the spectrum of aggressive B-cell lymphomas, characterised by unique clinical, immunophenotypic and molecular features. Since publication of the 2019 British Society for Haematology (BSH) Good Practice Paper, there have been important advances in imaging, radiotherapy approaches and systemic therapies, including chimeric antigen receptor (CAR) T-cell therapy and immune checkpoint inhibition. Key updates include the use of PET-adapted approaches to guide radiotherapy decisions following first-line chemoimmunotherapy, with omission of consolidative radiotherapy in patients achieving complete metabolic response. The guideline emphasises the importance of dose-intensive chemoimmunotherapy regimens and provides a framework for the clinical management of patients with Deauville stage 4 (DS4) and DS5 response status on PET-CT imaging following chemoimmunotherapy.

KEYWORDS
CAR T, chemoimmunotherapy, DLBCL, PMBCL, primary mediastinal B cell lymphoma

METHODOLOGY

This guideline was compiled according to the BSH process at [https://b-s-h.org.uk/media/16732/bsh-guidance-development-process-dec-5-18.pdf]. The Grading of Recommendations Assessment, Development and Evaluation (GRADE) nomenclature was used to evaluate levels of evidence and to assess the strength of recommendations. Recommendations are

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presented with both a strength[1–3] and quality of evidence (A–C), and recommendations were formulated through multidisciplinary expert consensus. The GRADE criteria can be found at http://www.gradeworkinggroup.org and is summarised in appendix 3 of the guidance document linked above.

Evidence was reviewed and synthesised by the guideline panel with a focus on clinically actionable outcomes. Recommendations were developed through iterative discussion and consensus among panel members. The strength of recommendations reflects both the quality and consistency of the available evidence and expert consensus in areas where high-quality evidence was limited. The wording of recommendations (e.g. 'we recommend', 'we suggest' or 'may be considered') was used to indicate the strength of recommendation and degree of supporting evidence.

LITERATURE REVIEW DETAILS

Recommendations are based on a structured review of published literature using MEDLINE (via PubMed). The search covered publications from January 2010 to December 2025 and was last updated in December 2025. Search criteria included the terms 'PMBCL', 'Primary mediastinal B-cell lymphoma', 'lymphoma', 'CAR-T', 'bispecific antibody' and 'radiotherapy', used in combination where appropriate. Reference lists of relevant articles were also screened to identify additional studies. Consideration was predominantly given to phase II and III clinical trials with early phase studies or case series only considered where there was a paucity of higher-grade evidence. Conference abstracts and non-English journals were predominantly excluded, although selected, high-relevance abstract data were considered where no full publication was available.

REVIEW OF THE MANUSCRIPT

Review of the manuscript was performed by the British Society for Haematology (BSH) Haemato-oncology Task Force and the BSH Guidelines Committee and the sounding board of BSH; these organisations do not necessarily approve or endorse the contents.

INTRODUCTION

Primary mediastinal B-cell lymphoma (PMBCL) is a mature, aggressive B-cell lymphoma considered to be of thymic B-cell origin, with distinctive clinical, immunophenotypic and molecular features. It typically arises in the anterior mediastinum with variable extension to the lung, pleura and pericardium. Regional lymph nodes can be involved, often the supraclavicular nodes. Bone marrow and extra-nodal involvement are uncommon at presentation but are more frequently seen at relapse.[1]

PRESENTATION

The age of presentation is typically in the third or fourth decade and there is a female predominance. Patients may present with symptoms due to pericardial or pleural effusions, a poor prognostic factor[2] or with compressive symptoms, including features of superior vena cava obstruction (SVCO).[3] Venous thromboembolism is common, particularly at the time of presentation. A retrospective analysis of 156 adults and children with PMBCL reported thrombotic complications in 28% of patients.[4] Although there is limited evidence, we recommend prophylactic anti-coagulation in view of this high thrombosis rate in patients with platelet counts $>50 \times 10^9$/L. Therapeutic dose anti-coagulation is advised for those with proven thrombosis. There is not enough evidence to recommend routinely giving therapeutic dose anti-coagulation to patients with venous compression and this should be determined on a case-by-case basis. A recent BSH guideline has been published on cancer-associated thrombosis and this should be referenced when making treatment decisions for these patients.[5]

DIAGNOSIS AND PATHOBIOLOGY OF PMBCL

The diagnosis of PMBCL can be challenging due to clinical and pathological similarities to classical Hodgkin lymphoma, mediastinal grey zone lymphoma and other B-cell lymphomas. A surgical biopsy may be required for some patients although a needle core biopsy is often adequate to make the histological diagnosis. If a core biopsy is performed, this should be of adequate gauge (14G) and if possible, at least two cores, accepting that there is a risk of tissue exhaustion after standard immunohistochemistry, potentially leaving insufficient material for fluorescence in situ hybridisation (FISH) or molecular testing. The clinicopathological features and management of each case should be reviewed in a multidisciplinary team meeting, which includes a specialist haematopathologist.[6]

Gene expression profiling has established PMBCL as a distinct B-cell lymphoma characterised by a signature that overlaps with classic Hodgkin lymphoma (cHL) and distinct from other aggressive lymphomas. The molecular hallmarks include deregulation of NF-kβ, JAK/STAT signalling pathways and mechanisms of immune escape.

Morphology and immunophenotype

PMBCL presents as a diffuse infiltration of neoplastic lymphoid cells distributed in sheets or clusters, usually in a fibrotic background. The neoplastic cells are usually medium sized to large with abundant pale cytoplasm and round or ovoid nuclei with small nucleoli. The lymphoma cells express CD45 and B-cell lineage antigens,

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for example, CD19, CD22, CD20 and CD79a, as well as transcription factors such as BOB1, OCT2 and PAX5. Most cases express CD30 (staining is typically weak and heterogeneous compared with cHL) and are usually IRF4 and CD23 positive with variable expression of BCL6 and BCL2. CD10 expression is uncommon (20%–30% of cases) and EBER (Epstein Barr Virus-encoded RNA) is negative. Occasional cases are CD15+ and the presence of remnants of thymic epithelium provide support for PMBCL when considering differential diagnoses. PDL1 and particularly PDL2 are typically expressed.[7]

Molecular tests—FISH and next-generation sequencing (NGS)

Certain cytogenetic abnormalities, for example, *CIITA* rearrangements, *REL* amplification and copy number abnormalities or rearrangements of the PD1 ligand genes, *PDL1* (*CD274*) and *PDL2* (*PDCD1LG2*), are generally specific to PMBCL although they can occur in other B-cell malignancies. PDL1 and PDL2 are located on chromosome 9p24.1 and 9p copy gain has been frequently described in PMBCL. *JAK2* is also upstream from PDL1 on 9p24.1, which further augments PD1 ligand expression in PMBCL.[8] High PDL1/PDL2 gene expression has been shown to correlate with worse outcome.[9] Other rearrangements commonly observed in diffuse large B-cell lymphoma (DLBCL), such as *BCL2*, *BCL6* and *MYC* rearrangements, are rare or absent. By NGS, the most commonly observed pathogenic variants occur in the genes *SOCS1* (60%–91%), *GNA13* (47%–53%), *STAT6* (33%–58%), *CD58* (20%–48%), *B2M* (36%–67%), *ITPKB* (40%–64%), *NFKBIE* (22%–49%), *TNFAIP3* (30%–60%) and *IL4R* (24%).[7,10]

Tests for PMBCL included on the National Health Service England National Cancer Genomic test directory are *REL* copy number, *PDL1* and *PDL2* rearrangement and copy number by FISH, or by a multitarget NGS panel.[11,12]

STAGING

It is recommended that a baseline positron emission tomography-computed tomography (PET-CT) scan is performed in all patients at diagnosis prior to corticosteroid treatment to ensure accurate assessment of disease extent, volume and degree of metabolic activity. Steroids should, however, not be withheld in clinical emergencies. CT staging is acceptable if waiting for a baseline PET introduces unacceptable delay in clinically urgent cases. Bone marrow involvement is seldom seen at diagnosis[2,13,14] and a bone marrow biopsy is not mandated although should be considered if there is concern regarding an alternative diagnosis. The blood tests required at presentation are the same as those described in the recently published BSH guidelines for the management of DLBCL.[15]

Consider contrast-enhanced CT at baseline especially when there is concern for vascular compromise or concomitant thromboembolic complications.

FERTILITY AND PREGNANCY

Fertility is particularly important in PMBCL given the typically young age of patients at presentation. Data from Gharwan et al.[16] demonstrated that although ovarian dysfunction occurred during chemotherapy, fertility was preserved in most patients; however, ovarian failure occurred in some patients over the age of 40.

Patients presenting with PMBCL who have reproductive potential should be offered referral to a reproductive specialist prior to chemoimmunotherapy. In cases where clinical urgency prevents review, patients should still be referred to a reproductive specialist for consideration of fertility assessment and preservation once chemoimmunotherapy has commenced. Consider assessment of gonadal function following treatment.

Sperm preservation should be offered to males before cancer directed treatment with testicular sperm extraction if unable to provide semen samples. For females, established fertility preservation methods should be offered including embryo, oocyte and ovarian tissue cryopreservation; the appropriate methods and timings of fertility preservation are outside of the scope of this guideline but are discussed in more detail in recent fertility guidelines.[17]

Pregnant patients should be managed in conjunction with obstetricians and anaesthetists specialising in high-risk pregnancies. Although chemotherapy in pregnancy carries significant risk to the mother and fetus (particularly in the first trimester), small retrospective studies have suggested that chemotherapy regimens that do not include antimetabolite agents can be safely given to pregnant patients after the second trimester.[18,19]

FIRST-LINE TREATMENT

First-line therapy for patients with newly diagnosed PMBCL is centred on dose- and time-intensive chemoimmunotherapy regimens including R-CHOP 14, R-ACVBP (rituximab, doxorubicin, cyclophosphamide, vindesine, bleomycin and prednisolone) and DA-R-EPOCH.

There is evidence that rituximab-containing dose-intensive regimens have superior efficacy to R-CHOP 21. A post hoc subgroup analysis of the United Kingdom National Cancer Research Institute (NCRI) Phase III R-CHOP 14 versus R-CHOP 21 prospective trial evaluated the outcomes for patients with PMBCL ($n = 50$).[20] Most patients (58%) received consolidative involved-site radiation therapy (ISRT) following R-CHOP. With a median follow-up of 7.2 years a 5-year progression-free survival (PFS) of 79.8% and 5-year overall survival (OS) of 83.8% were observed. In an exploratory

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analysis of this subgroup, there was a numerical but not statistically significant difference in event-free survival (EFS) between treatment arms, suggesting improved outcomes with the R-CHOP 14 approach, but the patient numbers were small.

Other studies have also confirmed superiority of dose-intensive rituximab-containing regimens (R-M/VACOPB and R-ACVBP) over R-CHOP 21.[21–24]

LYSA conducted a large multicentre retrospective analysis evaluating PMBCL patient outcomes following real-world first-line treatment for 313 patients. Patients received R-ACVBP ($n=180$), R-CHOP 14 ($N=76$) or R-CHOP 21 ($n=57$). Consolidation strategies varied with 25.6%, 31.6% and 1.8% receiving subsequent autologous haematopoietic stem cell transplant (AHSCT), respectively, and 5.4% overall receiving mediastinal radiotherapy. With a median follow-up of 44 months the 3-year PFS for R-ACVBP, R-CHOP 14 and R-CHOP 21 was 89.4%, 89.4% and 74.7% ($p=0.18$); and 3-year OS was 92.4%, 100% 87.5% respectively ($p=0.0036$).[21]

Sibon et al.[23] reported a pooled analysis of five prospective clinical trials of large B-cell lymphoma from the LYSA group evaluating 290 PMBCL patients. In this study 4-year PFS for patients treated with R-CHOP 21, R-CHOP 14 and R-ACVBP was 64%, 78% and 86% ($p=0.004$) and 4-year OS 73%, 89% and 88% ($p=0.031$) respectively, with treatment regimen remaining independently prognostic in multivariable analysis.

Evaluation of PMBCL regimens from Swedish registry data also indicated a relative survival benefit for patients treated with the more dose-intensive regimens of R-CHOP with etoposide administered every 14 days (R-CHOEP 14) and R-Hyper-CVAD (rituximab, cyclophosphamide, doxorubicin, vincristine, dexamethasone, methotrexate and cytarabine) versus R-CHOP.[22]

Further support for a dose-intensive regimen is shown in a meta-analysis of 4068 patients with PMBCL which demonstrated an 8% OS benefit for dose-intensive regimens (including DA-EPOCH-R $n=670$, R-CHOP 14/R-CHOEP 14 $n=199$, VACOP/MACOP-B +/− R $n=458$) versus CHOP +/− R 21 ($n=1551$). The survival benefit was only maintained when evaluating rituximab-containing regimens.[24]

The use of DA-R-EPOCH increased internationally following a small prospective phase II study where consolidative radiotherapy was omitted unless the patient had progressive disease. Dunleavy et al.[25] reported a prospective, non-randomised, single centre study of 6–8 cycles of DA-R-EPOCH in 51 patients with a further 16 patients reported from a retrospective second centre analysis. With a median follow-up of 5 years, the EFS was 93% and the OS was 97%. Complete remission (CR) rate was 96%, with two patients receiving radiotherapy for locally progressive disease. This cohort was followed up and added to, to include 93 patients.[26] Twenty-five patients had an end of treatment (EOT) scan reported as DS 4 (Deauville stage 4) or 5 (uptake 3 × liver uptake), 17 patients had a DS 4 response and only 1 had treatment failure. Of the eight patients who were DS 5, 4 had treatment failure. The authors concluded that the high rate of false positive EOT PET scans meant that a singular EOT PET did not accurately identify treatment failure, and serial PET imaging with decreasing SUV max discriminated post-treatment inflammatory change from residual disease. Retrospective analysis of DA-R-EPOCH used in children and adults has also demonstrated that consolidation radiotherapy is not required if an adequate PET response has been achieved.[4]

Although this Phase II study showed encouraging outcomes, the results are yet to be validated in a larger, multicentre randomised study for PMBCL patients. The CALGB phase III randomised trial compared R-CHOP with DA-R-EPOCH in previously untreated patients with DLBCL (only 6% with PMBCL). No difference in efficacy was demonstrated (including for the PMBCL cohort, $n=35$) but increased toxicity was observed with the DA-R-EPOCH.[27]

The IELSG37 trial randomised patients who were in a complete metabolic remission (CMR) after chemotherapy (DS 1–3) to receive radiotherapy consolidation or not. Patients were treated with the investigator's choice of rituximab-containing induction regimens R-CHOP-21 (12%), R-CHOP-14 (25%), R-V/MACOP-B (34%), DA-R-EPOCH (17%), other (12%). Patients treated with R-CHOP-21 showed a significantly higher percentage of DS 5 response compared to those treated with other regimens (23.8% vs. 8.2% average, $p<0.001$) and inferior EFS compared to all other dose-intensive regimens. The IELSG-37 trial was not powered to compare induction regimens, which were administered according to local practice. There was no difference in rates of CMR in patients who received R-CHOP-14 or DA-R-EPOCH and therefore either of these regimens are recommended. R-CHOP-21 is to be avoided where possible.[28,29]

Most patients will not require central nervous system (CNS) prophylaxis in view of their low CNS International prognostic index (IPI) score and low risk profile. The indications for CNS prophylaxis are the same as detailed in the BSH guidelines for de novo DLBCL.[30]

Recommendations

- **Offer participation in a clinical trial wherever possible (1A)**
- **Offer R-CHOP-14 × 6 or DA-R-EPOCH × 6 (1B)**
- **R-CHOP-21 is not recommended (1B)**
- **Thromboprophylaxis is recommended in view of the high thrombotic risk (1C)**
- **Anti-microbial prophylaxis is recommended as per local practice (1B)**

CHEMOIMMUNOTHERAPY TREATMENT RESPONSE ASSESSMENT

Interim imaging, either CT or PET-CT should be performed. Interim fluorodeoxyglucose-PET in PMBCL provides prognostic information but should not be used to guide treatment modification. Early metabolic response is associated

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with progression-free outcomes, with interim PET positivity linked to inferior survival; however, the positive predictive value remains limited, and there is insufficient evidence to support escalation or de-escalation of therapy based on interim PET findings alone.[31–33]

EOT PET-CT should be performed at 6 weeks following cycle 6 day 1 of treatment and reported using Deauville criteria (DS). If radiotherapy is subsequently delivered, then EOT response should be assessed with a 3-month post-radiotherapy PET-CT.[34] DS 1–3 represents CMR, in accordance with the Lugano Classification[35,36] and predicts an excellent prognosis following chemoimmunotherapy. DS 4 may represent inflammation or lymphoma. DS 5 is associated with poor prognosis and is usually due to lymphoma. In IELSG 37, DS 5 was calculated as uptake >2× liver. In the United Kingdom, many centres use >3× liver to allocate a DS 5 score.

In recent years, advanced PET reconstruction techniques have increased sensitivity and have been shown to alter Deauville scores in a proportion of cases.

Most scans in studies investigating PMBCL published to date have been reconstructed with conventional ordered subset expected maximisation (OSEM). Therefore, using reconstructions that match this (or specific research-accredited reconstructions) when stratifying treatment plans according to Deauville score is likely to be important while we await a new harmonisation performance standard to be established.

Lesion and liver SUV max should be reported to allow clinicians to assess the change in lesion uptake (Δ SUV max) especially in the Deauville 4/5 groups where a significant decrease in lymphoma activity in the D 4/5 group can be prognostic.[21]

Recommendations

- **Use conventional OSEM PET reconstruction techniques or specific research-accredited reconstructions for reporting the Deauville score on response assessment scans (1A)**
- **There are no current data to support a change of treatment as a result of interim PET response; therefore, the modality of interim imaging should be as per local clinical practice to assess response (2B)**

RADIOTHERAPY CONSOLIDATION AFTER FIRST-LINE CHEMOIMMUNOTHERAPY

The limited efficacy of second-line chemotherapy emphasises the importance of first-line treatment for maximising cure rates. The progress in radiotherapy (RT) techniques has also mitigated against immediate toxicities and late effects.[37–40] Despite these advances in radiotherapy, routine consolidation RT for all patients would lead to significant overtreatment. The high negative predictive value (NPV) of EOT PET-CT has led several groups to adopt PET-guided consolidative RT strategies[41,42] and the utility of this strategy was confirmed for DS 1–3 (CMR) patients by the IELSG37 trial.

Results from the IELSG37 trial indicate favourable outcomes in patients who achieved CMR, with no survival detriment for patients randomised to observation only (omission of consolidation RT). A total of 268 (49.2%) patients of the 545 patients determined to be in CMR by central review were randomised to either observation ($n = 132$) or RT ($n = 136$). The 30-month PFS rates were 96.2% and 98.5% respectively, with a stratified hazard ratio of 1.47 (95% confidence interval [CI], 0.34 to 6.28) and absolute risk difference of 0.68% (95% CI, −0.97% to 7.46%). The 5-year OS was 99% across both study arms.[29]

For the 50.8% of patients not in a CMR after first-line chemoimmunotherapy, those with EOT DS 4 had significantly better outcomes than those with DS 5. Patients with DS 4 had a PFS, which was not significantly different to those in CMR (95.8% and 97.5% respectively). By contrast, patients with DS 5 experienced significantly inferior 5-year PFS and OS (60.3% and 74.6% respectively). The excellent outcomes of DS 4 patients were observed in the context of consolidation RT delivered to 86% of patients. It remains unknown what proportion of patients with DS 4 represented residual PMBCL versus post-chemotherapy inflammatory tissue. However, with modern RT techniques and the smaller RT volume needed after chemoimmunotherapy, the potentially improved local disease control needs to be balanced against the potential late RT toxicity in individual case assessments.

The optimal approach to patients who achieve DS 4 after first-line chemoimmunotherapy remains controversial. A DS4 response is heterogeneous; for some patients, the SUV is just above liver and others significantly higher, and this is an important point that needs discussing and reviewing in the multidisciplinary team (MDT). Some advocate for consolidative RT,[43] while others propose a more conservative strategy that involves ongoing surveillance by repeating a PET scan in a maximum of 4 to 6 weeks. An ongoing reduction in SUV on surveillance PET scan may indicate that radiotherapy could be omitted. In a study by Melani et al., 17 patients treated with DA-R-EPOCH had DS 4 and only 1 progressed following serial surveillance.[26] In a small cohort from the Royal Marsden, only two of nine patients with a DS 4 after R-CHOP-21 experienced progressive disease (PD).[44] Notwithstanding, it is important to not delay RT if there is concern about active disease and risk compromising the excellent outcome seen after RT in IELSG37.

In view of the controversy regarding the optimal management of patients achieving a DS 4, a personalised strategy dependent on the age, sex and site of disease/residual mass and lesion SUV max compared to liver SUV max should be considered with the clinical (radiation) oncologist in consultation with the patient and after careful MDT discussion. If possible, a biopsy should be performed. For most patients, RT will be feasible with low risk of late toxicity (using cardiac and breast sparing techniques), if only the residual PET-positive areas are treated. In selected cases, where there is only minimal uptake above the liver uptake and higher

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doses to the heart tissue or female breast cannot be avoided, a watch-and-wait strategy with interval PET-CT imaging to monitor metabolic evolution or diffusion-weighted (DW) magnetic resonance imaging is a reasonable option. If repeat imaging does not show significant improvement, a biopsy should be sought and RT should be considered to avoid the risk of progression with further delay. The psychological impact of serial scanning and the associated uncertainty should also be considered.

For patients with DS 5 at EOT, a re-biopsy should be performed wherever possible. For most (those with evidence of progression or significant areas of DS 5 activity on PET-CT) patients, systemic treatment as for relapsed PMBCL is recommended. Consolidation RT can be considered if there is limited single residual uptake on PET-CT without progression or if the patient is unfit for systemic chemotherapy. RT should be provided using the best technique for individual patients, usually using deep inspiration breath hold (DIBH) photons or protons in selected cases.

Recommendations

- **Do not offer radiotherapy for patients with DS 1–3 following chemoimmunotherapy (1A)**
- **Re-biopsy is recommended if feasible for all DS 4 and DS 5 patients at the EOT (1B)**
- **MDT discussion is essential to review the following (1B):**
  - **The heterogeneity of the DS 4 group—there is more concern regarding patients with SUV nearly 2× that of liver compared to just above liver**
  - **The site of disease and risk of radiotherapy**
  - **Patient factors such as age and sex**
- **A personalised approach to balance the risk versus benefit of surveillance PET-CT or radiotherapy consolidation is required for patients with DS 4 at EOT (2C)**
- **Patients with DS 4 at EOT should be offered consultation with a clinical (radiation) oncologist to discuss radiotherapy versus surveillance imaging (2A)**
- **Discuss all patients with DS 5 at EOT at an MDT and perform a biopsy if possible**
  - **Offer systemic treatment as for relapsed disease for patients with DS 5 (1B)**
  - **Radiotherapy alone can be considered if there is limited single residual uptake on PET-CT without progression or if the patient is unfit for systemic chemotherapy (2B)**

RELAPSED PMBCL

Relapsed disease is relatively rare for patients with PMBCL but when it occurs, response rates with second-line high-dose chemotherapy regimens are lower than for patients with DLBCL.[45] When relapse occurs, this is almost always within 1-year[46] and the rare cases of late relapse often occur in the setting of CNS relapse.[2,47–49] Please see Figures 1 and 2 for suggested treatment algorithms for relapsed PMBCL.

Chimeric antigen receptor (CAR) T-cell treatment

Lisocabtagene maraleucel (Liso-cel) is licensed in a second-line setting and axicabtagene ciloleucel (axi-cel) and liso-cel are licensed in a third-line setting.

The TRANSFORM study[50] randomised patients who were refractory or relapsed within 12 months of first-line chemotherapy to liso-cel or standard second-line chemotherapy and AHSCT. Eighteen patients with PMBCL were enrolled in total with 10 in the standard of care arm. While outcomes for this subgroup were not described separately, in the standard of care arm, 6 of the 10 had crossed over to liso-cel as planned per protocol underscoring the poor outcomes seen with second-line chemotherapy for these patients. Liso-cel is licensed and NICE (National Institute for Health and Care Excellence) approved for patients with PMBCL who relapse within 12 months of first-line treatment. Axi-cel is not approved as a second-line treatment for PMBCL patients due to their exclusion from the ZUMA 7 trial.[51]

In a third-line setting both axi-cel and liso-cel are licensed options in the United Kingdom. ZUMA 1 led to the licensing of axi-cel for third-line treatment of DLBCL. Eight of 101 patients infused had PMBCL although the outcomes for this cohort were not analysed separately. Three of the eight patients remain 'at risk' in the 5-year follow-up suggesting there are long-term remissions.[52,53]

The phase 1/2 TRANSCEND study investigated Liso-cel for patients with DLBCL in a third-line setting and patients with PMBCL were eligible. Fifteen patients (6%) were enrolled onto the study and 14 were evaluable with an overall response rate (ORR) of 79% (95% CI 49–95%) with a 50% CR rate (95% CI 23–77). Median duration of response and PFS were not reached in this subgroup compared to 5.6 and 3 months for those with DLBCL.[54]

Real-world data for CAR T-cell therapy in PMBCL

Real-world data show encouraging efficacy for patients with PMBCL and are summarised in Table 1.

Consolidation autologous transplant

Patients with PMBCL often have a poor response to second-line high-dose chemotherapy and are therefore unable to proceed to AHSCT.[45]

European Society for Blood and Marrow Transplantation reported the outcomes of 86 patients who underwent

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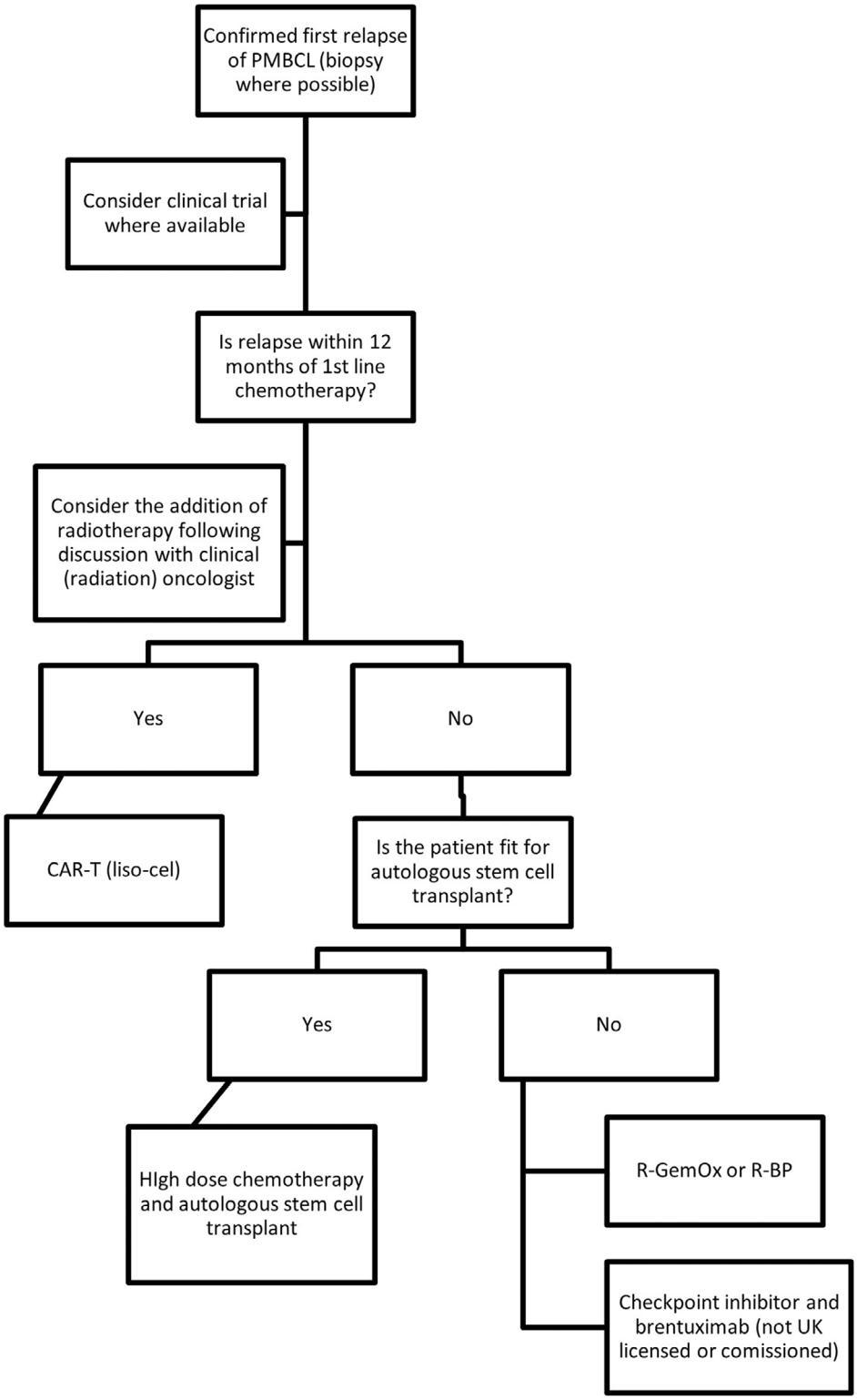


FIGURE 1 Algorithm for the treatment of patients with first relapse of PMBCL. RBP, rituximab, bendamustine, polatuzumab; R-GemOx, rituximab, gemcitabine and oxaliplatin.

autologous transplant between 2000 and 2012. Sixteen were performed as consolidation following first-line treatment. Forty-four patients were transplanted with chemosensitive disease having achieved CR/partial response (PR) and 24 were chemorefractory (stable disease/progressive disease) at the point of the transplant. 3-year PFS and OS for those with chemosensitive disease were 94% and 100% compared to 39% and 41% for those with chemoresistant disease.

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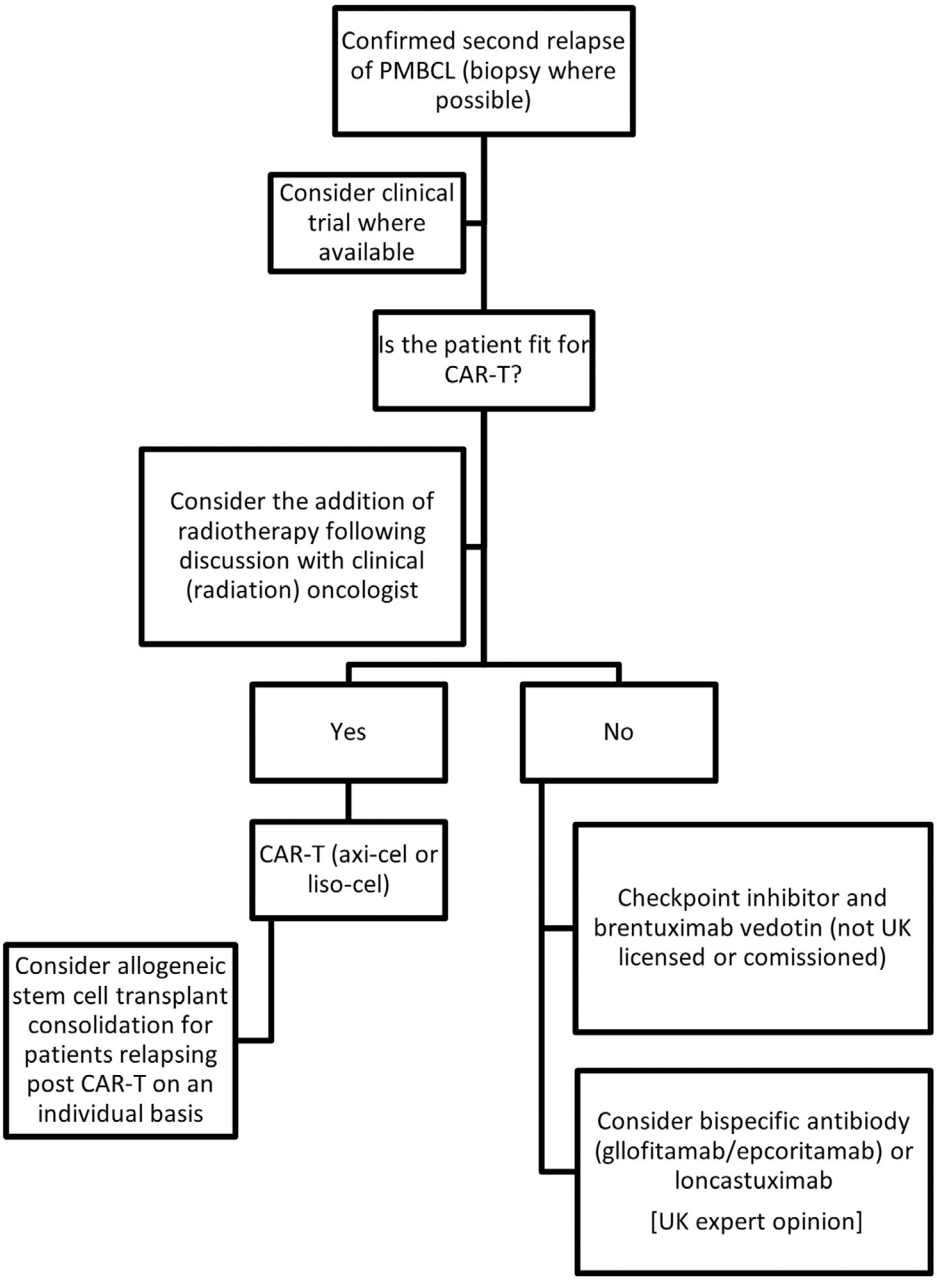


FIGURE 2 Algorithm for the treatment of second or further relapse of PMBCL. RBP, rituximab, bendamustine, polatuzumab treatment of patients with second or further relapse of PMBCL; R-GemOx, rituximab, gemcitabine and oxaliplatin.

Thirty-five per cent of those in the chemorefractory group had radiotherapy post-AHSCT.[55] This highlights that patients with chemosensitive disease can benefit from autologous transplant but that patients with PMBCL are often chemorefractory in a second-line setting.

Fifty-one patients undergoing AHSCT between 1989 and 2014 in the Memorial Sloan Kettering Cancer Center (MSKCC) were reported (42% relapsed and 58% primary refractory). ORR with high dose chemotherapy was 65% in 60 patients. Fifty-two patients also received radiotherapy as part of their treatment strategy with a median dose of 30 Gy in those who had details available. Fifty-one of the sixty proceeded to transplant; 75% had chemosensitive disease going into transplant. The 3-year PFS was 60% and OS 65%.[56]

The MD Anderson Cancer Center reported the outcomes for the 60 patients in their centre who underwent AHSCT between 2000 and 2022. In patients responding to second-line chemotherapy, the 5-year PFS in 60 patients who underwent autologous transplant was 58%, although 30% of patients had received consolidation radiotherapy post-transplant.[57]

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TABLE 1 Real-world data for outcomes of CAR T-cell therapy in PMBCL.

Study	Study design	Results
CART-SIE {Chiappella, 2025, CART-SIE REAL LIFE STUDY: PRIMARY MEDIASTINAL B-CELL LYMPHOMA (PMBCL) HAVE A SUPERIOR OUTCOME COMPARED TO LARGE B-CELL LYMPHOMA (LBCL) TREATED WITH AXICABTAGENE CILOLEUCEL}	Prospective observational study of patients receiving CAR T-cell therapy in Italy	Improved PFS and OS in 70 patients with PMBCL compared to 190 with LBCL (unmatched groups) CR rates similar to LBCL (50%)
Real-world outcomes of axicabtagene ciloleucel in adult patients with primary mediastinal B-cell lymphoma[50]	Retrospective	33 patients with PMBCL, ORR 70% and CR 69%. 24-month PFS 64%
Outcomes of axicabtagene ciloleucel in PMBCL compare favourably with those in DLBCL: a GLA/DRST registry study.[51]	Retrospective	13 patients with PMBCL, ORR 85%, CR 54%
DESCAR-T Registry {Galtier, 2025, Outcomes of patients with relapsed or refractory primary mediastinal B-cell lymphoma treated with anti-CD19 CAR-T cells: CARTHYM, a study from the French national DESCAR-T registry}	Retrospective	82 patients with PMBCL received CAR T-cell therapy. 68% CR, 24-month PFS 71% in 76 patients who received axi-cel.
CIBMTR analysis (US series 2025) {Gauthier, 2025, CD19 CAR T-Cell Therapy for Primary Mediastinal Large B-Cell Lymphoma: A CIBMTR Analysis}	Retrospective	135 patients with PMBCL received CAR T-cell therapy. 67% CR, 24-month PFS 58.6%.

Radiotherapy

With increasing evidence that radiotherapy may be safely omitted in those, having initial therapy for PMBCL,[29] the role of radiotherapy in the relapsed/refractory (R/R) setting is increasing. Supportive evidence is largely anecdotal, for example, two patients treated with radiotherapy alone at the time of relapse remained in remission.[25] One meta-analysis of 12 studies exploring the role of radiotherapy in the R/R setting showed, despite predominantly low-quality studies, that radiotherapy for PMBCL led to similar outcomes compared to DLBCL with a 2-year PFS and OS for PMBCL being 95% and 97%.[58] Recommendations from expert consensus groups also affirm the value of radiotherapy in the R/R setting.[59]

The timing, RT dose and volume for patients with biopsy-proven refractory or relapsed PMBCL should otherwise follow the same principles as those for other aggressive non-Hodgkin lymphomas and should be given in accordance with International Lymphoma Radiation Oncology Group (ILROG) guidelines.[60] Post-transplant RT is preferred because of the pneumonitis risk when RT precedes high-dose therapy. Butterfly intensity-modulated RT or volumetric-modulated arc therapy and proton beam therapy,[61] along with DIBH, are encouraged to limit the doses to the lungs, heart and breast tissue.

Data for PMBCL in recent pivotal trials for novel agents in R/R DLBCL

Loncastuximab is licensed and commissioned in the United Kingdom for use for DLBCL including PMBCL for third line onwards following the Lotis 2 trial. Eligibility criteria included PMBCL but excluded those with bulky disease. Seven patients (5%) of patients had PMBCL. While the ORR and PFS were 49% and 4.5 months respectively, only one patient of the seven with PMBCL achieved a PR with a PFS of 1.35 months.[62]

Bispecific antibodies are drugs that activate endogenous immune cells by co-targeting tumour antigens and T or natural killer cells. Anti-CD3 × CD20 bispecifics have demonstrated remarkable activity in patients with heavily pretreated B-cell lymphomas. Both glofitamab and epcoritamab are bispecific antibodies that are licensed and commissioned in the United Kingdom for use third line and beyond for R/R DLBCL. Both pivotal trials allowed patients with PMBCL to enrol onto the study. The EPCORE NHL-1 trial enrolled four patients with PMCBL but did not detail their outcomes separately from the full cohort.[63,64] The phase 2 study that led to the licensing of glofitamab included six patients with PMBCL in the overall cohort the primary end-point, CR was 39% (95% CI 32–48) and for the PMBCL subgroup was 50% (95% CI 12–88).[65] Bispecific combination data are encouraging but more data are required: two patients treated with glofitamab and polatuzumab achieved a CMR with a 5-month follow-up.[66]

Allogeneic transplant in PMBCL

Registry data demonstrate that allogeneic transplant can lead to durable remissions. Of 33 patients in a Belgian and French review, the median PFS and OS were respectively 47% (95% CI 33%–68%) and 48% (95% CI 33%–70%) with disease status at time of transplant identified as a strong predictor for durable remissions.[67] A multicentre US retrospective review of 28 patients who underwent allogeneic transplant showed 5-year PFS of 34% and OS 45%. Again, those with chemosensitive disease at the time of transplant had significantly better outcomes with 5-year PFS of 50% and OS 58%.[68]

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Brentuximab single agent

Despite the expression of CD30, a phase 2 study showed only 2/15 patients with relapsed PMBCL had a response with brentuximab with no CRs.[69]

Checkpoint inhibitors

Pembrolizumab as a single agent for up to 2 years treatment has been investigated in phase 1b study Keynote 013 and the follow-on phase 2 studies Keynote 170 and Keynote A33 in a cohort of seven Japanese patients. Among the 21 patients in the KN013 study, the ORR was 48% and CR rate 33% and in the follow-on KN170, the ORR was 45% with CR rate 13% in 53 patients. None of those patients who achieved a CR subsequently relapsed by time of study reporting.[70] With longer follow-up of over 4 years in the KN170 study, CR rate was 21% (some PRs converted to CRs) and seven patients had consolidated their response with a transplant. Median PFS for the whole cohort was 4.3 months.[71] The Japanese cohort of seven patients had an ORR of 43%.[72]

The combination of nivolumab and brentuximab has been explored in the Checkmate 436 study. Thirty patients with R/R PMBCL were treated until disease progression or toxicity. ORR was 73% with a CR rate of 40% as per independent review. Twelve patients proceeded to consolidation transplant, although four patients remained in long-term remission without consolidation. Median PFS was 26 months (95% CI 2.6–not reached).[71,73] These favourable outcomes have also been replicated in real-world analysis of 14 patients.[74]

There are case reports of nivolumab being used in conjunction with radiotherapy for R/R PMBCL[75] and high dose chemotherapy.[76]

Recommendations for the management of relapsed PMBCL

- **Offer participation in a clinical trial wherever possible (1A)**
- **Offer CD19-directed targeted chimeric antigen receptor (CAR) T-cell (CAR-T) therapy (liso-cel) as second-line treatment if the patient has relapsed within 12 months (1C)**
- **Offer CD19-directed CAR T-cell therapy (axi-cel or liso-cel) as third or subsequent treatment line (1C)**
- **If available, consider the unlicensed combination of a PD1 inhibitor with brentuximab vedotin for patients with disease progression following CAR-T or autologous transplant (2C)**
- **Consider radiotherapy to localised sites of chemotherapy refractory disease (2C)**
- **Consider CD20-directed bispecific antibody with epcoritamab or glofitamab as a third or subsequent line of treatment line (2C)**
- **Consider loncastuximab as a third or subsequent line of treatment (2C)**
- **Consider allogeneic transplant for occasional patients who have failed multiple lines of treatment including CAR-T (2C)**

Follow-up

As the majority of relapses occur within 12 months, by 24 months most patients who remain in remission may be considered for discharge back to primary care. Patients should only be re-scanned if they have new clinical symptoms or signs suggestive of relapse.[34]

Patients should be aware of potential long-term complications (for example, cardiac toxicity and second malignancies) so that prompt referral can be made if symptoms arise.

Recommendations

- **Consider discharge back to primary care for patients who are 24 months following end of treatment and who remain in remission (2C).**
- **Potential long-term complications should be outlined to patients and primary care health care professionals.**
- **Do not perform surveillance imaging in asymptomatic patients (1B).**

AUTHOR CONTRIBUTIONS

Cathy Burton: Conceptualization; methodology; writing – original draft; writing – review and editing. **Wendy Osborne:** Conceptualization; writing – original draft; methodology; writing – review and editing; supervision. **Kate Cwynarski:** Conceptualization; writing – original draft; methodology; writing – review and editing. **Hannah Bielby:** Writing – original draft; writing – review and editing; methodology; conceptualization. **N. George Mikhaeel:** Conceptualization; methodology; writing – original draft; writing – review and editing. **Timothy Illidge:** Conceptualization; methodology; writing – original draft; writing – review and editing. **Amos Burke:** Conceptualization; writing – original draft; methodology; writing – review and editing. **Andrew Davies:** Conceptualization; writing – original draft; methodology; writing – review and editing; supervision. **Mary Gleeson:** Conceptualization; writing – original draft; methodology; writing – review and editing. **George Petrides:** Conceptualization; writing – original draft; methodology; writing – review and editing. **Dima El-Sharkawi:** Conceptualization; writing – original draft; methodology; writing – review and editing. **Christopher P. Fox:** Conceptualization; writing – original draft; methodology; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

The BSH paid the expenses incurred during the writing of this guidance. All authors have made a declaration of interests to the BSH, which may be viewed on request.

DATA AVAILABILITY STATEMENT

Only peer reviewed published data are included within this guideline, all referenced in the guideline.

REVIEW PROCESS

Members of the writing group will inform the writing group chair if any new evidence becomes available that would alter the strength of the recommendations made in this document or render it obsolete. The document will be reviewed regularly by the relevant task force and the literature search will be re-run every 5 years to search systematically for any new evidence that may have been missed. The document will be archived and removed from the BSH current guidelines website if it becomes obsolete. Please check the BSH guidelines website (www.b-s-h.org.uk/guidelines) for any addenda that may be produced after the initial publication.

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